

Piyush Kaswa

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SUMMARY

Results-driven **Data Engineer** skilled in building scalable data pipelines using **Python, SQL, PySpark**, and **AWS**, with strong foundations in **data structures and ETL workflows**. Experienced in delivering reliable analytics platforms on **AWS Glue and Redshift**, while actively building expertise in **Generative AI, LLMs, RAG pipelines, and AI agents**. Comfortable developing **Python services**, integrating **REST APIs**, and working with **Airflow, Terraform, and MLOps/CI-CD concepts**.

WORK EXPERIENCE

Data Engineer — Startappss Systems India Pvt. Ltd

May 2025 – Present

- Developed a hybrid data ingestion framework integrating structured and semi-structured data from on-premises databases, APIs, Amazon S3, and Kafka into a unified data lake and warehouse.
- Implemented Medallion Architecture (Bronze, Silver, Gold layers) using Amazon S3 (Parquet) and Amazon Redshift for scalable, modular data pipelines.
- Engineered Apache Airflow DAGs to automate and schedule complex ETL workflows, reducing manual intervention.
- Applied Slowly Changing Dimension (SCD) Type 2 logic to maintain historical tracking of sales data, supporting time-series analytics.
- Built real-time streaming pipelines using Amazon MSK (Kafka) and AWS Lambda for low-latency ingestion.
- Designed Amazon Athena querying layers on S3 for fast, cost-efficient exploration without additional ETL overhead.
- Automated infrastructure provisioning with Terraform for environment consistency and scalability across AWS regions.

PROJECTS

Data Ingestion & Migration Pipeline — Willis Towers Watson (WTW)

- Developed a hybrid data ingestion framework integrating door-to-door sales data (CSV on S3), online sales data (on-prem SQL Server & RDS), and Kafka streams via AWS Lambda.
- Implemented Bronze/Silver/Gold Medallion Architecture using Amazon S3 (Parquet) and Redshift for analytics-ready storage.
- Built SCD Type 2 tracking in the Silver layer to maintain historical integrity; created Athena views for ad-hoc querying.
- Built Kafka-based real-time ingestion pipelines and migrated streaming data to Redshift's raw layer for downstream processing.
- Automated infrastructure with Terraform for reproducible deployments across AWS regions.
- Optimized ETL workflows using AWS Lambda for lightweight transformations and efficient data hand-offs.
- Delivered a unified pipeline that improved throughput, reduced latency, and enabled faster data-driven decision-making.

LLM-Powered Document Intelligence & RAG System — Personal Project

- Designed and built a production-ready **Retrieval-Augmented Generation (RAG)** pipeline to enable accurate question answering over large collections of unstructured documents.
- Implemented data ingestion and preprocessing workflows to clean, chunk, and embed documents using **Python**, supporting scalable retrieval from vector storage.
- Developed prompt workflows and LLM-based reasoning logic to generate grounded, context-aware responses with reduced hallucinations.
- Built backend services to expose the RAG system via **REST APIs**, enabling seamless integration with downstream applications.
- Applied efficient data structures and indexing strategies to optimize retrieval latency and improve response relevance at scale.
- Implemented logging, evaluation, and basic monitoring to debug model behavior and improve system reliability.
- Documented system design, experiments, and learnings to ensure maintainability and knowledge transfer.

EDUCATION

Acropolis Institute of Technology and Research, Indore 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 7.14

- Relevant Coursework: Specialization in Artificial Intelligence and Machine Learning

ACHIEVEMENTS & CERTIFICATIONS

- **NPTEL Certification** — The Joy of Computing using Python
- **IBM (Coursera)** — Python for Data Science, AI, and Development

SKILLS

Languages & Libraries:	Python, PySpark, SQL, NumPy, Pandas
AI / ML & GenAI:	Machine Learning, Image Classification, Generative AI, Prompt Engineering, RAG (Retrieval-Augmented Generation), LLM Concepts
Data Warehousing & Databases:	PostgreSQL, Snowflake, Amazon Redshift, Hive, Azure SQL Database
Data Engineering Tools & Platforms:	Apache Airflow, Apache Kafka, Dagster, AWS Glue, AWS Lambda, AWS S3, Hadoop
Orchestration & IaC:	Apache Airflow, Terraform
Data Formats & Querying:	Parquet, JSON, Amazon Athena
Cloud & MLOps (Exposure):	AWS, CI/CD Concepts, Model Deployment, Monitoring